Soil Health & Crop Recommendation Report

Report Date: November 23, 2025 at 18:40:57 UTC

Farmer Information	
Name:	test 1
Phone:	8088117817
Location:	Vijayapura, Vijayapura taluk
Land Size:	1.00 acres

Parameter	Value	Unit
Nitrogen (N)	38.0	mg/kg
Phosphorus (P)	61.0	mg/kg
Potassium (K)	21.0	mg/kg
pH Level	6.50	
Temperature	30.9	°C
Humidity	67.0	%
Rainfall	119.0	mm
Soil Type	Black Soil	

Soil Health Assessment

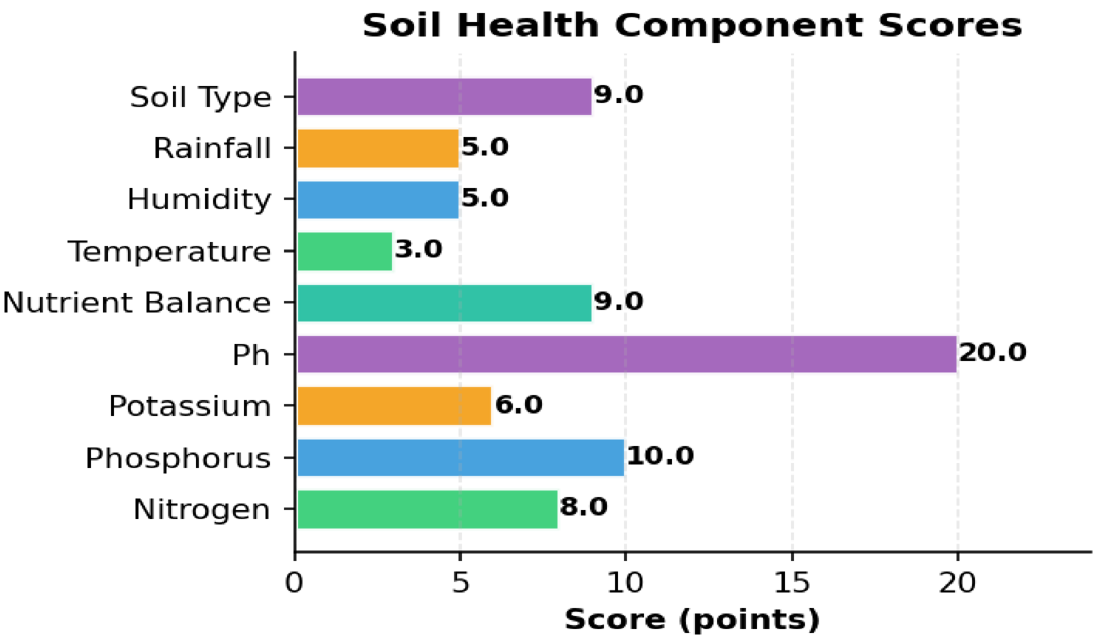
Overall Soil Health Score	75.0/100	Good

Component	Score	Points
Nitrogen	8.0	points
Phosphorus	10.0	points
Potassium	6.0	points
Ph	20.0	points
Nutrient Balance	9.0	points
Temperature	3.0	points
Humidity	5.0	points
Rainfall	5.0	points

Soil Health & Crop Recommendation System

■ Powered by ML + IoT + Weather Intelligence

Soil Type	9.0	points



Weather Suitability for Pigeonpeas

Overall Suitability Score: 7.3/10 ■ Good

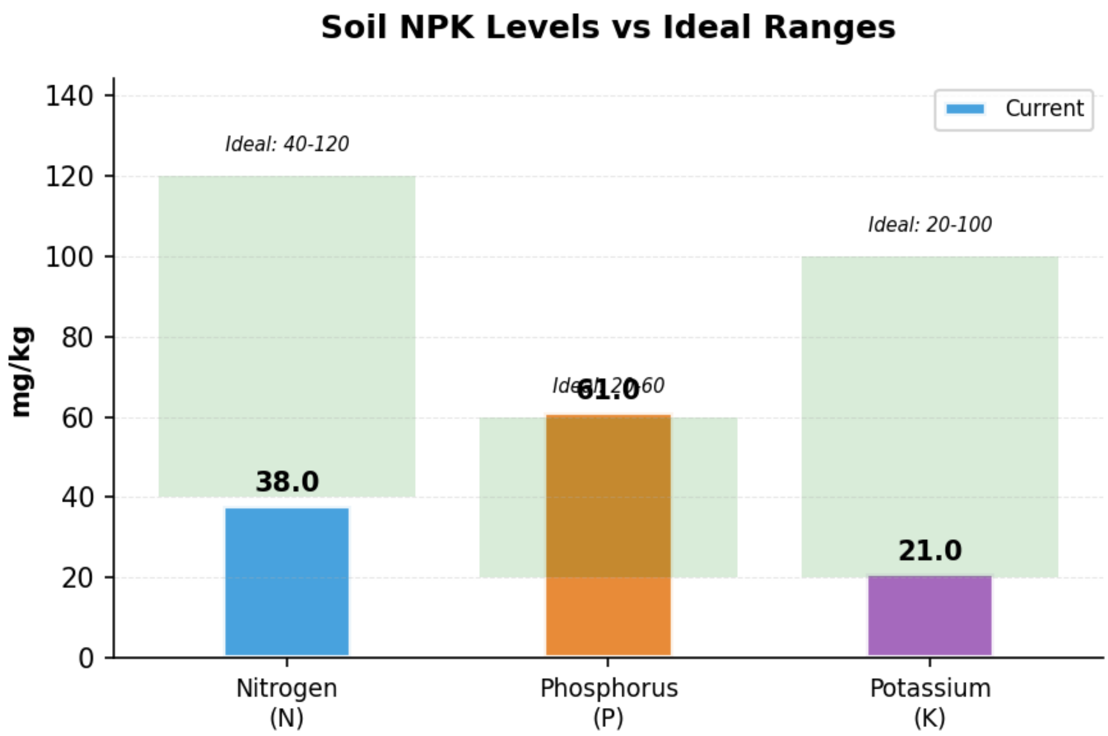
Weather conditions are good for this crop

- ✓ Temperature 30.9°C is ideal for pigeonpeas
- ✓ Humidity 67.0% is ideal for pigeonpeas — reduces disease risk
- ■ Rainfall 119 mm is below requirement (600-1000 mm); irrigation strongly recommended

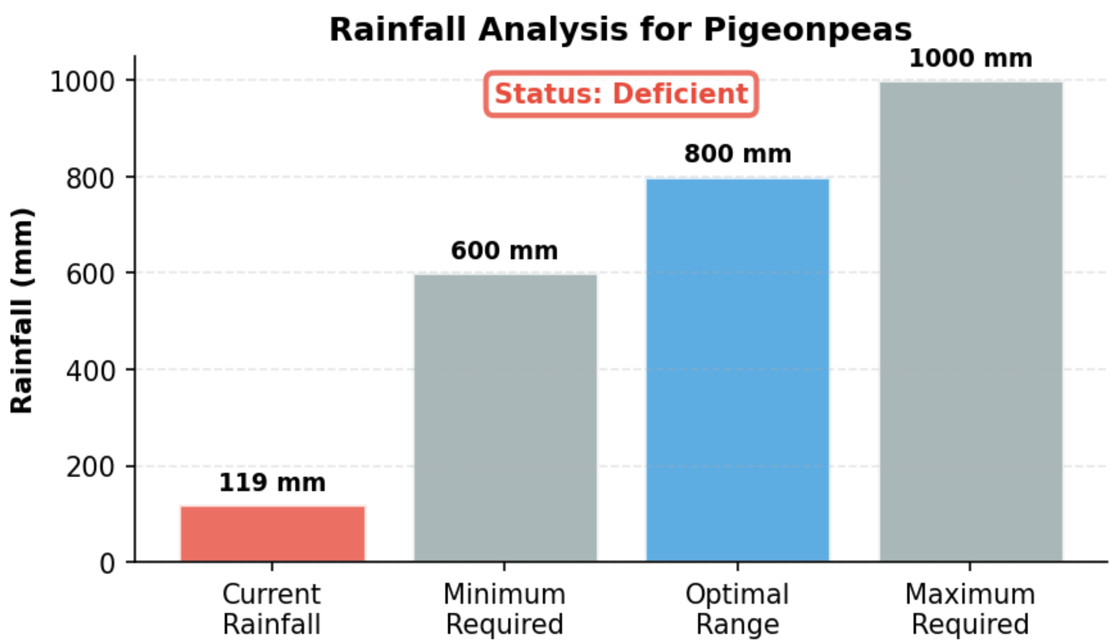
Recommended Crop

Predicted Crop	Pigeonpeas
Confidence Level	94.6%
Model Used	Stacking Classifier

Soil NPK Levels Visualization



Rainfall Analysis



■■ Warnings & Observations

Action Required:

- ■ Nitrogen low → Apply compost or urea

- ■ Phosphorus excessive → May interfere with micronutrients
- ■ Rainfall below optimal → Increase irrigation

Positive Indicators:

- ✓ Potassium level optimal
- ✓ pH optimal (6.5)
- ✓ Soil type: Black Soil

■ Market Price & Profitability Analysis

Price Range	2000 - 3500 ■/Quintal
Average Price	2750 ■/Quintal

Best Market: Local Market, Vijayapura taluk (2750 ■/Quintal)

Market Advice: Market data integration is available in the system. In a production environment, this would show real-time market prices.

■ Soil Treatment Recommendations

- Soil pH (6.5) is within optimal range (6.0-7.5) for pigeonpeas.
- Nitrogen (38.0 mg/kg) is excessive for pigeonpeas. Optimal range: 10-30. Reduce nitrogen application to avoid excessive vegetative growth.
- Phosphorus (61.0 mg/kg) is excessive for pigeonpeas. Optimal range: 40-60. High P can interfere with micronutrient uptake.
- Potassium (21.0 mg/kg) is within optimal range for pigeonpeas.
- High clay content, retains moisture well. Ideal for cotton, wheat, and pulses.

■ Fertilizer Recommendations

FARM DETAILS

Land Size: 1.0 acres

Current Soil NPK: N=38.0 | P=61.0 | K=21.0 mg/kg

Optimal NPK for pigeonpeas: N=20 | P=50 | K=20 kg/acre

SOIL NUTRIENT STATUS

✓ Excellent: Soil NPK levels are already optimal for pigeonpeas

Maintain soil health through:

- Regular organic compost application
- Crop rotation practices
- Green manuring between seasons

ORGANIC & BIO-FERTILIZER ALTERNATIVES

Farm Yard Manure (FYM): 5-10 tons

■■ Application: 5-10 tons/acre (provides balanced NPK)

Compost: Well-decomposed organic compost

■■ Application: 3-5 tons total

Green Manure: Dhaincha, Sunhemp, or other leguminous crops

■■ Benefits: Natural nitrogen fixation, improves soil structure

Bio-Fertilizers:

- Rhizobium: 200g/acre (for nitrogen fixation)
- PSB (Phosphate Solubilizing Bacteria): 200g/acre
- Azospirillum: 200g/acre (for nitrogen fixation)
- VAM (Vesicular Arbuscular Mycorrhiza): Enhances P uptake

Note: Apply fertilizers based on soil test results and local
agricultural extension recommendations for best results.

■ Crop-Specific Tips for Pigeonpeas

Drought-tolerant legume. Requires well-drained soil and warm climate.

■ Additional Notes & Disclaimer

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.